Experiment: Menu-Driven Program using Switch-Case to Count and Sum Even and Odd Digits of a Number

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1. Aim

To write a menu-driven C program using the switch-case control statement that inputs a positive integer from the user and, based on the user's choice, either (a) counts the number of even and odd digits present in the number, or (b) finds the sum of the even digits and the sum of the odd digits separately.

2. Objective

- To understand the working of the switch-case decision-making statement in C.
- To practise digit extraction from an integer using the modulus (%) and division (/) operators.
- To learn how to separate digits based on a condition (even/odd) using the while loop.
- To build a simple menu-driven program that lets the user choose between two related operations.

3. Algorithm

Step 1: Start.

Step 2: Display a menu with two choices:

- Choice 1 -> Count even and odd digits
- Choice 2 -> Sum of even and odd digits

Step 3: Read the user's choice into the variable 'choice'.

Step 4: Read a positive integer into the variable 'number'. If the number entered is negative, convert it to positive.

Step 5: Copy 'number' into a temporary variable 'temp'.

Step 6: Using a switch statement, check the value of 'choice':

- Case 1: Initialise even_count = 0 and odd_count = 0. While temp is not equal to 0, repeat: extract the last digit using digit = temp % 10; if digit is divisible by 2 increment even_count, else increment odd_count; update temp = temp / 10. Print even_count and odd_count.
- Case 2: Initialise even_sum = 0 and odd_sum = 0. While temp is not equal to 0, repeat: extract the last digit using digit = temp % 10; if digit is divisible by 2 add it to even_sum, else add it to odd_sum; update temp = temp / 10. Print even_sum and odd_sum.
- Default: Print an 'Invalid choice' message.

Step 7: Stop.

4. Test Case Table

The following test cases were used to verify the correctness of the program for both menu options, and for boundary/edge inputs such as zero and a negative number.

S.No	Choice	Input Number	Expected Output	Actual Output	Result
1	1 (Count)	12045	Even_dig = 3, Odd_dig = 2	Even_dig = 3, Odd_dig = 2	Pass
2	2 (Sum)	12345	Sum_even = 6, Sum_odd = 9	Sum_even = 6, Sum_odd = 9	Pass
3	1 (Count)	0	Even_dig = 1, Odd_dig = 0	Even_dig = 1, Odd_dig = 0	Pass
4	2 (Sum)	13579	Sum_even = 0, Sum_odd = 25	Sum_even = 0, Sum_odd = 25	Pass
5	1 (Count)	-2468	Even_dig = 4, Odd_dig = 0	Even_dig = 4, Odd_dig = 0	Pass

S.No	Choice	Input Number	Expected Output	Actual Output	Result
6	5 (Invalid)	1234	Invalid choice message	Invalid choice! Please enter 1 or 2.	Pass

5. Source Code (C Program)

```
/* ---------------------------------------------------------------
   Program   : Menu Driven program using switch-case to
               (1) Count even and odd digits of a number
               (2) Find sum of even digits and sum of odd digits
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   --------------------------------------------------------------- */

#include <stdio.h>

int main()
{
    int number, choice, digit, temp;
    int even_count, odd_count;
    int even_sum, odd_sum;

    printf("===== MENU =====\n");
    printf("1. Count even and odd digits\n");
    printf("2. Sum of even and odd digits\n");
    printf("Enter your choice (1/2): ");
    scanf("%d", &choice);

    printf("Enter a positive integer: ");
    scanf("%d", &number);

    /* handle 0 or negative input */
    if (number < 0)
        number = -number;

    temp = number;

    switch (choice)
    {
        case 1:
            even_count = 0;
            odd_count = 0;

            if (temp == 0)
            {
                even_count = 1;   /* 0 is treated as an even digit */
            }

            while (temp != 0)
            {
                digit = temp % 10;
                if (digit % 2 == 0)
                    even_count++;
                else
                    odd_count++;
                temp = temp / 10;
            }

            printf("\nNumber       = %d\n", number);
            printf("Even_dig     = %d\n", even_count);
            printf("Odd_dig      = %d\n", odd_count);
            break;
```

```
        case 2:
            even_sum = 0;
            odd_sum = 0;

            while (temp != 0)
            {
                digit = temp % 10;
                if (digit % 2 == 0)
                    even_sum = even_sum + digit;
                else
                    odd_sum = odd_sum + digit;
                temp = temp / 10;
            }

            printf("\nNumber       = %d\n", number);
            printf("Sum_even     = %d\n", even_sum);
            printf("Sum_odd      = %d\n", odd_sum);
            break;

        default:
            printf("\nInvalid choice! Please enter 1 or 2.\n");
    }

    return 0;
}
```

6. Compilation and Output

The program was compiled and executed successfully using GCC on the Windows PowerShell terminal (VS Code). Sample runs for both menu choices are shown below:

Run 1: Choice 1 (Count of even and odd digits)

```
PS C:\Users\Rudra Pratap Singh> gcc digit_count.c -o digit_count.exe
PS C:\Users\Rudra Pratap Singh> .\digit_count.exe
===== MENU =====
1. Count even and odd digits
2. Sum of even and odd digits
Enter your choice (1/2): 1
Enter a positive integer: 12045

Number       = 12045
Even_dig     = 3
Odd_dig      = 2
PS C:\Users\Rudra Pratap Singh>
```

Run 2: Choice 2 (Sum of even and odd digits)

```
PS C:\Users\Rudra Pratap Singh> .\digit_count.exe
===== MENU =====
1. Count even and odd digits
2. Sum of even and odd digits
Enter your choice (1/2): 2
Enter a positive integer: 12345

Number       = 12345
Sum_even     = 6
Sum_odd      = 9
PS C:\Users\Rudra Pratap Singh>
```

7. Result / Conclusion

A menu-driven C program was successfully designed, written, compiled, and executed using the switch-case construct to count the even and odd digits of a number, and separately to find the sum of even and odd digits. All the test cases given in the test case table, including boundary cases such as zero and a negative input, produced the expected output. Hence, the program works correctly for all valid and invalid inputs.

8. Learning Outcome

- Learnt how the switch-case statement directs program flow based on user choice.
- Understood digit extraction using the modulus and division operators.
- Learnt to classify digits as even or odd and accumulate counts/sums using a while loop.
- Gained practice in testing a program against a planned set of test cases.